INMO(28th) – 2013

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Let Γ_1 and Γ_2 be two circles touching each other externally at R. Let l_1 be a line which is tangent to Γ_2 at P and passing through the centre O_1 of Γ_1. Similarly let l_2 be a line which is tangent to Γ_1 at Q and passing through the centre O_2 of Γ_2. Suppose l_1 and l_2 are not parallel and intersect at K. If $KP = KQ$, prove that the triangle PQR is equilateral.
2. Find all positive integers m, n and primes $p \geq 5$ such that $m(4m^2 + +12) = 3(p^n - 1)$.
3. Let a, b, c, d be positive integers such that $a \geq b \geq c \geq d$. Prove that the equation $x^4 - ax^3 - bx^2 - cx - d = 0$ has no integer solution.
4. Let n be a positive integer. Call a nonempty subset S of {1, 2, 3, ….,n} good if the arithmetic mean of the elements of S is also an integer. Further let t_n denotes the number of good subsets of {1, 2, 3, ….,n}. Prove that t_n and n both odd or both even.
5. In an acute triangle ABC, O is the circumcentre, H is the orthocenter and G is the centroid. Let OD be perpendicular to BC and HE be the perpendicular to CA, with D on BC and E on CA. Let F be the mid point of AB, suppose the areas of triangles ODC, HEA and GFB are equal. Find all possible values of $\angle C$.
6. Let a, b, c, x, y, z be positive real numbers such that $a + b + c = x + y + z$ and $abc = xyz$. Further suppose that $a \leq x < y < z \leq c$ and $a < b < c$. Prove that $a = x, b = y, c = z$.

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